

MLFCS

Vector Spaces: Span, Linear (in)dependence & Basis

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Today's plan

- ▶ Recall: Span of a set of vectors
- ▶ Recall: Linear (in)dependence of a set of vectors
- ▶ Definition: Basis of a vector space
- ▶ Our go-to examples of two-tuples or three-tuples of rationals
 - ▶ (standard) basis for these vector spaces
 - ▶ size of basis for these vector spaces
 - ▶ inner product for these vector spaces
 - ▶ orthogonal basis for these vector spaces

} Last
lecture

← New



Vector space V over a field F → Two operations $+$ & $\times$

Let V be a set of vectors and F be any field. Then V is said to be a vector space over the field F if the following conditions hold:

For any vectors $\vec{u}, \vec{v}, \vec{w} \in V$ and any scalars $r, s \in F$

- (1) Commutativity of vector addition: $\vec{u} \oplus \vec{v} = \vec{v} \oplus \vec{u}$
 - (2) Associativity of vector addition: $(\vec{u} \oplus \vec{v}) \oplus \vec{w} = \vec{u} \oplus (\vec{v} \oplus \vec{w})$
 - (3) Existence of Additive identity: $\vec{0} \oplus \vec{v} = \vec{v}$
 - (4) Existence of additive inverse: for each $\vec{x}$, there exists $\vec{-x}$ such that $\vec{x} \oplus \vec{-x} = \vec{0}$
 - (5) Associativity of multiplication of scalar & vector: $r(s\vec{v}) = (rs)\vec{v}$
 - (6) Distributivity of scalar sums: $(r + s)\vec{v} = r\vec{v} \oplus s\vec{v}$
 - (7) Distributivity of vector sums: $r(\vec{u} \oplus \vec{v}) = r\vec{u} \oplus r\vec{v}$
 - (8) Existence of identity of multiplication of scalar & vector: $1\vec{v} = \vec{v}$
- multiplication of a scalar with a vector*

First, we need to define two operations for above 8 conditions to make sense:

- ▶ Vector addition: for each $\vec{u}, \vec{v}$ a vector from V is assigned to $\vec{u} \oplus \vec{v}$
- ▶ Multiplication of a scalar by a vector: for each $s \in F$ and $\vec{v} \in V$, a vector from V is assigned to $s\vec{v}$

Note that multiplication of two vectors is not defined here!



Go-to examples of a vector space

The set of 2-tuples of rational numbers is a vector space over the rational numbers:

- ▶ The set of 2-tuples of rational numbers is defined as $\mathbb{Q}^2 := \left\{ \begin{pmatrix} a \\ b \end{pmatrix} : a, b \in \mathbb{Q} \right\}$

$\mathbb{Q}^2$

The set of 3-tuples of rational numbers is a vector space over the rational numbers:

$\mathbb{Q}^3$

- ▶ The set of 3-tuples of rational numbers is defined as

$$\mathbb{Q}^3 := \left\{ \begin{pmatrix} a \\ b \\ c \end{pmatrix} : a, b, c \in \mathbb{Q} \right\} \rightarrow \text{for every choice of } a, b, c \in \mathbb{Q}$$

- ▶ To verify each of the 8 conditions for the above two examples to be vector spaces, you will just need to use the fact that $\mathbb{Q}$ is a field.

- ▶ Details for first example given in solution of Q4 of Week 8 exercise sheet



Span of a set of vectors

- ▶ Let V be a vector space over a field F .
- ▶ Let $\vec{v}_1, \vec{v}_2$ be two vectors in V
- ▶ We define $\text{Span}(\vec{v}_1, \vec{v}_2) := \{r_1 \vec{v}_1 \oplus r_2 \vec{v}_2 \mid r_1, r_2 \in F\}$
 - ▶ All possible linear combinations of $\vec{v}_1$ and $\vec{v}_2$
 - ▶ Span of $\vec{v}_1$ and $\vec{v}_2$
 - $\vec{v}_1 \in \text{Span}(\vec{v}_1, \vec{v}_2)$ by choosing $r_1 = 1$ & $r_2 = 0$
 - $\vec{v}_2 \in \text{Span}(\vec{v}_1, \vec{v}_2)$ by choosing $r_1 = 0$ & $r_2 = 1$
- ▶ Exercise: $\text{Span}(\vec{v}_1, \vec{v}_2)$ is a vector space over the field F .
 - ▶ Simple verification of 8 conditions & using the fact that V is a vector space over F ←
 - ▶ Details presented in solution of Q3 in Week 9 exercise sheet

In general, given any set of vectors from V we can define its span.

$$\hookrightarrow \text{Span}(\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4) \dots$$



How to check if a given vector belongs to a span?

Consider the vector space $\mathbb{Q}^3$ over $\mathbb{Q}$ of 3-tuples of rational numbers.

► Consider the vectors $\vec{v}_1 = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}$, $\vec{v}_2 = \begin{pmatrix} 5 \\ 6 \\ 10 \end{pmatrix}$, $\vec{v}_3 = \begin{pmatrix} 11 \\ 300 \\ 14 \end{pmatrix}$ from $\mathbb{Q}^3$

► I want to check if the vector $\vec{w} = \begin{pmatrix} 41 \\ 12 \\ 110 \end{pmatrix}$ belongs to $\text{Span}(\vec{v}_1, \vec{v}_2, \vec{v}_3)$

► How can we do that?

► Recall that $\text{Span}(\vec{v}_1, \vec{v}_2, \vec{v}_3) := \left\{ r_1 \vec{v}_1 \oplus r_2 \vec{v}_2 \oplus r_3 \vec{v}_3 \mid r_1, r_2, r_3 \in F \right\}$ $\rightarrow F = \mathbb{Q}$

$\vec{w} \in \text{Span}(\vec{v}_1, \vec{v}_2, \vec{v}_3)$ if $\exists r_1, r_2, r_3 \in \mathbb{Q}$ s.t. $\vec{w} = r_1 \vec{v}_1 \oplus r_2 \vec{v}_2 \oplus r_3 \vec{v}_3$

$$\begin{pmatrix} 41 \\ 12 \\ 110 \end{pmatrix} = r_1 \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} + r_2 \begin{pmatrix} 5 \\ 6 \\ 10 \end{pmatrix} + r_3 \begin{pmatrix} 11 \\ 300 \\ 14 \end{pmatrix} \rightarrow \begin{array}{l} 41 = r_1 + 5r_2 + 11r_3 \\ 12 = 3r_1 + 6r_2 + 300r_3 \\ 110 = 4r_1 + 10r_2 + 14r_3 \end{array} \quad \begin{array}{l} \text{--- ①} \\ \text{--- ②} \\ \text{--- ③} \end{array}$$

solve ①, ②, ③ by Gaussian elimination to get

$$r_1 = \frac{3232}{245} \quad r_2 = \frac{1479}{245} \quad r_3 = \frac{-52}{245}$$

See solutions of Q1 and Q2 of Week 9 exercise sheet for more examples about spans



Linear (in)dependence of a set of vectors

- ▶ Let V be a vector space over a field F .
- ▶ A set of vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n \in V$ is **linearly independent** if $r_1 \vec{v}_1 \oplus r_2 \vec{v}_2 \oplus \dots \oplus r_n \vec{v}_n = \vec{0}$ implies $r_1 = r_2 = \dots = r_n = 0$

- ▶ Otherwise, the set of vectors is said to be linearly dependent

→ All of $r_1, r_2, \dots, r_n$ should be 0

- ▶ Does $\vec{0}$ always belongs to the span of any set of vectors?

- ▶ Yes, because $0 \vec{v} = \vec{0} \quad \forall \vec{v} \in V$

- ▶ Therefore, a set of vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n \in V$ is linearly independent if the **only way** to obtain $\vec{0}$ in its span is by taking all the scalars to be 0

- ▶ Question: Can $\vec{0}$ belong to any set of linearly independent vectors?

- ▶ No, because $1 \cdot 0 \cdot \vec{0} = \vec{0}$

- ▶ Next slide: How can we check if a given set of vectors is linearly independent or not?



Checking linear independence of a set of vectors

Consider the vector space $\mathbb{Q}^3$ over $\mathbb{Q}$ of 3-tuples of rational numbers.

► Consider the vectors $\vec{v}_1 = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}$, $\vec{v}_2 = \begin{pmatrix} 5 \\ 6 \\ 10 \end{pmatrix}$, $\vec{v}_3 = \begin{pmatrix} 11 \\ 300 \\ 14 \end{pmatrix}$ from $\mathbb{Q}^3$

► Want to check if these three vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are linearly independent

► How can we do that?

► let $r_1, r_2, r_3 \in \mathbb{Q}$ s.t. $r_1 \vec{v}_1 \oplus r_2 \vec{v}_2 \oplus r_3 \vec{v}_3 = \vec{0}$

$$r_1 \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} + r_2 \begin{pmatrix} 5 \\ 6 \\ 10 \end{pmatrix} + r_3 \begin{pmatrix} 11 \\ 300 \\ 14 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

a) Either the only solution is $r_1 = r_2 = r_3 = 0$. Then these vectors are lin. independent $\leftarrow$ G.E

b) Otherwise, lin. dependent if there is atleast one solution with not all of r_1, r_2, r_3 being 0

$$r_1 + 5r_2 + 11r_3 = 0$$

$$3r_1 + 6r_2 + 300r_3 = 0$$

$$4r_1 + 10r_2 + 14r_3 = 0$$

$\Rightarrow$ Always has $r_1 = r_2 = r_3 = 0$ as one solution

See solution of Q4 of Week 9 exercise sheet for how to show two given vectors in the vector space of 2-tuples of rational numbers are not linearly independent



Basis of a vector space

(Definition)

Let V be a vector space over a field F .

Definition: A set of vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n \in V$ forms a **basis** if **both** the following two conditions are satisfied:

▶ $\text{Span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n) = V$

▶ That is, every vector in V can be represented as a linear combination of the given vectors

▶ If $\text{Span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n) = V$ then the set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is called as a **spanning** set

▶ The set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is **linearly independent**

▶ That is, $r_1 \vec{v}_1 \oplus r_2 \vec{v}_2 \oplus \dots \oplus r_n \vec{v}_n = \vec{0}$ implies $r_1 = r_2 = \dots = r_n = 0$

Consider our go-to examples of the vector spaces:

▶ $\mathbb{Q}^2$ over $\mathbb{Q}$, i.e., 2-tuples of rational numbers over the field of rational numbers

▶ $\mathbb{Q}^3$ over $\mathbb{Q}$, i.e., 3-tuples of rational numbers over the field of rational numbers

Two questions about these (go-to) examples of vector spaces:

▶ Do these vector spaces have a basis? $\rightarrow$ Yes

▶ If yes, can you find one? $\rightarrow$ Yes



$\mathbb{Q}^2$ over $\mathbb{Q}$.

Basis for vector space of 2-tuples of rational numbers over the field of rational numbers

Definition: A set of vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n \in V$ forms a **basis** if **both** the following two conditions are satisfied:

► $\text{Span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n) = V$

$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ form a ~~standard~~ basis

► That is, every vector in V can be represented as a linear combination of the given vectors

► If $\text{Span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n) = V$ then the set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is called as a **spanning** set

► The set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is linearly independent

► That is, $r_1 \vec{v}_1 \oplus r_2 \vec{v}_2 \oplus \dots \oplus r_n \vec{v}_n = \vec{0}$ implies $r_1 = r_2 = \dots = r_n = 0$

• Spanning: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ & $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are spanning set

$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ & $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ can span every vector of $\mathbb{Q}^2$

Consider any general vector $\begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{Q}^2$

$$\begin{pmatrix} a \\ b \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\star)$$

• Linear Independence: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ & $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are linearly independent

$$\text{Suppose } \exists r_1, r_2 \text{ s.t. } r_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + r_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$r_1 \cdot 1 + r_2 \cdot 0 = 0 \Rightarrow r_1 = 0$$

$$r_1 \cdot 0 + r_2 \cdot 1 = 0 \Rightarrow r_2 = 0$$



Basis for vector space of 3-tuples of rational numbers over the field of rational numbers

Definition: A set of vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n \in V$ forms a **basis** if **both** the following two conditions are satisfied:

- ▶ $\text{Span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n) = V \rightarrow$ standard basis is $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$
- ▶ That is, every vector in V can be represented as a linear combination of the given vectors
- ▶ If $\text{Span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n) = V$ then the set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is called as a **spanning** set
- ▶ The set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ is linearly independent
- ▶ That is, $r_1 \vec{v}_1 \oplus r_2 \vec{v}_2 \oplus \dots \oplus r_n \vec{v}_n = \vec{0}$ implies $r_1 = r_2 = \dots = r_n = 0$

• Spanning set

Consider any vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbb{Q}^3$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \oplus b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \oplus c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

• Linearly independent

Suppose $\exists r_1, r_2, r_3$ s.t.

$$r_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \oplus r_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \oplus r_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow r_1 = 0 = r_2 = r_3$$

Two theorems about basis of vector spaces

Two theorems about basis of vector spaces

- ▶ Theorem 1: Every vector space has a basis
- ▶ Theorem 2: Every basis of a vector space has the same number of vectors
 - ▶ The number of vectors in a basis of a vector space is called as its dimension

Proofs of Theorem 1 and Theorem 2 are beyond the scope of this module!

Questions:

- ▶ What is the dimension of the vector space $\mathbb{Q}^2$ of two-tuples of rational numbers over the field of rational numbers?
 - ▶ Answer: 2 *One basis is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ & $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$*
- ▶ What is the dimension of the vector space $\mathbb{Q}^3$ of three-tuples of rational numbers over the field of rational numbers?
 - ▶ Answer: 3 *→ because one basis is $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$*



Definition: Inner product of two vectors

Recall (slide 4 of this lecture) that **we did not define** multiplication of two vectors for (general) vector spaces!

However, we can define (inner) product of two vectors for our two go-to examples of vector spaces:

- ▶ Vector space $\mathbb{Q}^2$ of two-tuples of rational numbers over the field of rational numbers $\mathbb{Q}$
 - ▶ If $\vec{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, then the inner product of $\vec{u}$ and $\vec{v}$ is defined as $\vec{u} \cdot \vec{v} = (u_1 \times v_1) + (u_2 \times v_2)$
 - ▶ Example: $\begin{pmatrix} 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 6 \end{pmatrix} = (1 \times 2) + (3 \times 6) = 20$
- ▶ Vector space $\mathbb{Q}^3$ of ~~two~~^{three} tuples of rational numbers over the field of rational numbers $\mathbb{Q}$
 - ▶ If $\vec{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$, then the inner product of $\vec{u}$ and $\vec{v}$ is defined as $\vec{u} \cdot \vec{v} = (u_1 \times v_1) + (u_2 \times v_2) + (u_3 \times v_3)$
 - ▶ Example: $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = (1 \times 4) + (2 \times 5) + (3 \times 6) = 4 + 10 + 18 = 32$



Definition: Orthogonal basis

We can define what it means for a basis to be orthogonal if there is some notion of inner product between vectors

Hence, we can define **orthogonal** basis for our two go-to examples of vector spaces:

- ▶ Vector space $\mathbb{Q}^2$ of two-tuples of rational numbers over the field of rational numbers $\mathbb{Q}$

- ▶ A basis $\{\vec{u}, \vec{v}\}$ for $\mathbb{Q}^2$ over $\mathbb{Q}$ is orthogonal if $\vec{u} \cdot \vec{v} = 0$

- ▶ Question: Does this vector space have an orthogonal basis?

↳ Yes, because standard basis $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is orthogonal

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (1 \times 0) + (0 \times 1) = 0$$

- ▶ Vector space $\mathbb{Q}^3$ of ~~two~~_{three}-tuples of rational numbers over the field of rational numbers $\mathbb{Q}$

- ▶ A basis $\{\vec{u}, \vec{v}, \vec{w}\}$ for $\mathbb{Q}^3$ over $\mathbb{Q}$ is orthogonal if $\vec{u} \cdot \vec{v} = 0$, $\vec{u} \cdot \vec{w} = 0$ and

$\vec{v} \cdot \vec{w} = 0$ All three inner products must be 0 for basis to be orthogonal

- ▶ Question: Does this vector space have an orthogonal basis?

↳ Yes, because the standard basis $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ is orthogonal

↳ Verify



What is the advantage of an orthogonal basis?

Consider the vector space $\mathbb{Q}^2$ of two-tuples of rational numbers over the field of rational numbers $\mathbb{Q}$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = (1 \times 1) + (1 \times -1) = 0$$

► Consider the orthogonal basis given by $\vec{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is orthogonal

► I want to find how $\vec{w} = \begin{pmatrix} 11 \\ -9 \end{pmatrix}$ can be represented as linear combination of

► Since $\{\vec{u}, \vec{v}\}$ form a basis we know $\vec{w} \in \text{Span}(\vec{u}, \vec{v})$

► Suppose $\vec{w} = r\vec{u} \oplus s\vec{v}$. How can we find r and s easily?

$$\begin{pmatrix} 11 \\ -9 \end{pmatrix} = r \begin{pmatrix} 1 \\ 1 \end{pmatrix} \oplus s \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{--- ①}$$

To find r , take inner product with $\vec{u}$ on both sides of ①

$$r = \frac{\vec{u} \cdot \vec{w}}{\vec{u} \cdot \vec{u}} = \frac{2}{2} = 1$$
$$\vec{u} \cdot \vec{w} = r \vec{u} \cdot \vec{u} + \underbrace{s \vec{u} \cdot \vec{v}}_0$$

To find s , take inner product with $\vec{v}$ on both sides of ①

$$s = \frac{\vec{v} \cdot \vec{w}}{\vec{v} \cdot \vec{v}}$$


Summary of the lecture

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 - ▶ Recall: Span of a set of vectors
 - ▶ Recall: Linear (in)dependence of a set of vectors

- ▶ Definition: Basis of a vector space ←
- ▶ Our go-to examples of two-tuples or three-tuples of rationals
 - ▶ (standard) basis for these vector spaces
 - ▶ size of basis for these vector spaces
 - ▶ inner product for these vector spaces ← multiplication of two vectors
 - ▶ orthogonal basis for these vector spaces

